

Time Series Stock Forecast

Libraries

```
In [1]: !date
```

Tue Apr 14 15:42:09 CDT 2026

```
In [113... # load required modules
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import dates as mdates
from prophet import Prophet
# import fix_yahoo_finance as yf
import yfinance as yf
from pandas_datareader import data as pdr
import math
from sklearn.metrics import mean_absolute_error
from prophet.plot import plot_plotly, plot_components_plotly
# import plotly.graph_objs as go # this can be solved by installing '!pip in
```

Selecting a company with the start and end date of its stock prices.

download dataset from Yahoo based on the user input (company, start date, end date)

```
In [114... stock = 'TSLA'
start = "2017-01-01"
end = "2023-12-08"
# df = pdr.get_data_yahoo(stock, start, end)
df = yf.download(stock, start, end)
dfa=df
```

```
[*****100%*****] 1 of 1 completed
```

```
In [115... df.shape
```

```
Out[115... (1745, 5)
```

```
In [116... df.head()
```

Out [116...

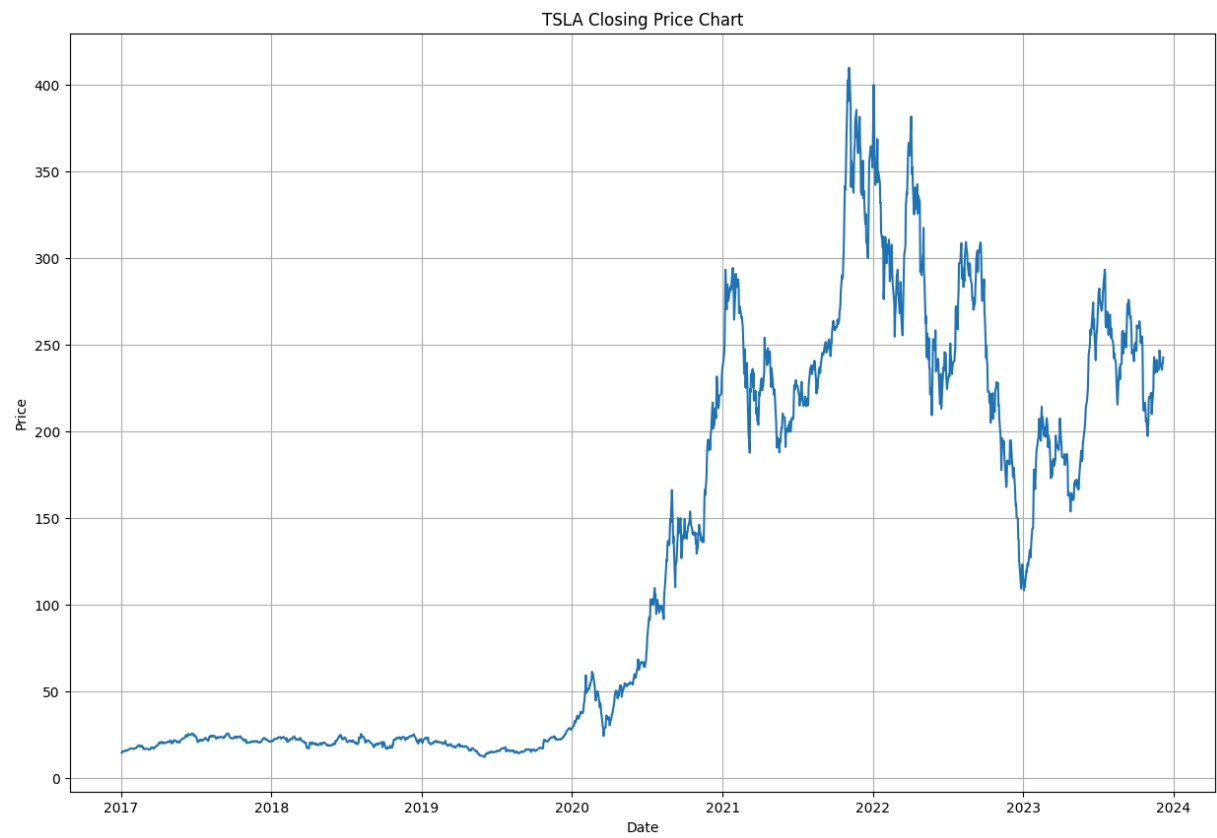
Price	Close	High	Low	Open	Volume
Ticker	TSLA	TSLA	TSLA	TSLA	TSLA
Date					
2017-01-03	14.466000	14.688667	14.064000	14.324000	88849500
2017-01-04	15.132667	15.200000	14.287333	14.316667	168202500
2017-01-05	15.116667	15.165333	14.796667	15.094667	88675500
2017-01-06	15.267333	15.354000	15.030000	15.128667	82918500
2017-01-09	15.418667	15.461333	15.200000	15.264667	59692500

Closing Price

In [117...

```
# df['Adj Close'].plot(figsize=(12,8))
# plt.show()

plt.figure(figsize=(15,10))
plt.plot(df['Close'])
plt.title(stock + ' Closing Price Chart')
plt.xlabel('Date')
plt.ylabel('Price')
plt.grid(True)
plt.show()
```



```
In [118... print(df.columns)
```

```
MultiIndex([( 'Close', 'TSLA'),  
            ( 'High', 'TSLA'),  
            ( 'Low', 'TSLA'),  
            ( 'Open', 'TSLA'),  
            ('Volume', 'TSLA')],  
           names=['Price', 'Ticker'])
```

High and Low prices

```
In [119... #fig, ax = plt.subplots(figsize=(16, 4))  
#(df.groupby(pd.Grouper(freq='W'))[['Low', 'High']]).mean().plot(color=['Red'
```

```
In [120... df.columns = df.columns.get_level_values(0)
```

```
In [121... df.head()
```

```
Out[121...      Price    Close    High    Low    Open    Volume  
      Date  
2017-01-03  14.466000  14.688667  14.064000  14.324000  88849500  
2017-01-04  15.132667  15.200000  14.287333  14.316667  168202500  
2017-01-05  15.116667  15.165333  14.796667  15.094667  88675500  
2017-01-06  15.267333  15.354000  15.030000  15.128667  82918500  
2017-01-09  15.418667  15.461333  15.200000  15.264667  59692500
```

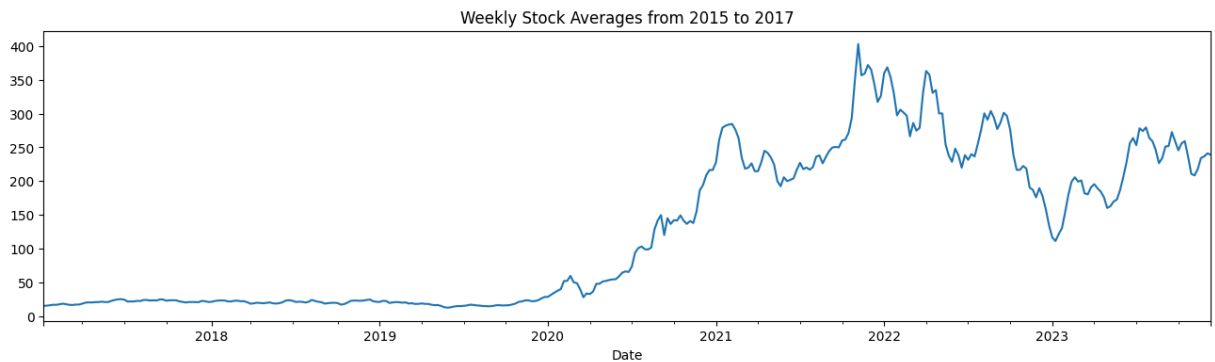
```
In [122... #fig, ax = plt.subplots(figsize=(16, 4))  
#df.groupby(pd.Grouper(freq='W'))[['Low', 'High']].mean().plot(  
    #ax=ax,  
    #title=stock + ' High and Low Price'  
#)  
  
#plt.show()
```

Weekly Stock Analysis Average

```
In [123... #weekly_avg = df.resample('W').mean()  
#weekly_avg
```

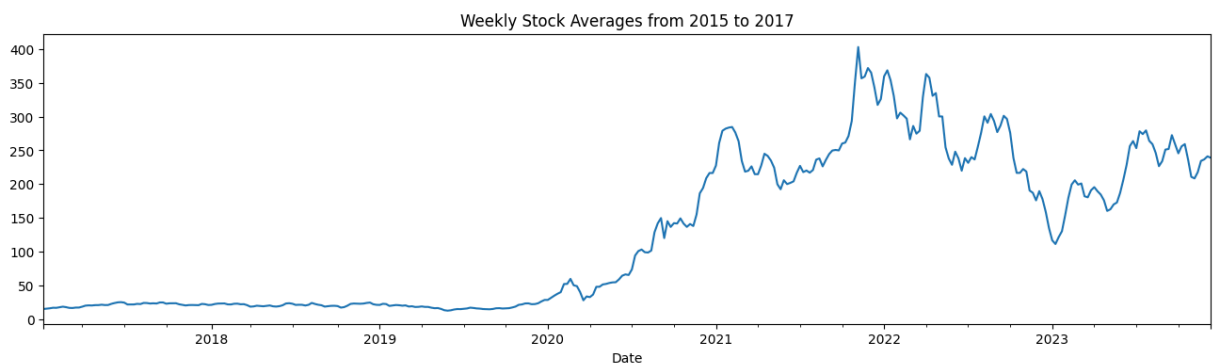
```
In [124... fig, ax = plt.subplots(figsize=(16, 4))  
  
weekly_avg['Close'].plot(  
    title='Weekly Stock Averages from 2015 to 2017',  
    ax=ax  
)
```

```
plt.show()
```



```
In [125... fig, ax = plt.subplots(figsize=(16, 4))
weekly_avg['Close'].plot(title='Weekly Stock Averages from 2015 to 2017', ax
```

```
Out[125... <Axes: title={'center': 'Weekly Stock Averages from 2015 to 2017'}, xlabel
='Date'>
```



Monthly Returns

Create new column 'Returns', 'Month_Name',
Month_name_Year', and 'Month'

for compute closing price each month

```
In [126... monthly = df.asfreq('BM')
monthly['Returns'] = df['Close'].pct_change().dropna()

monthly['Month_Name'] = monthly.index.strftime("%b")
monthly['Month_Name_Year'] = monthly.index.strftime("%b-%Y")

monthly = monthly.reset_index()
monthly['Month'] = monthly["Date"].dt.month
```

```
/var/folders/g7/8fm0cc6d7p13pn53xrhqdw5c0000gn/T/ipykernel_33896/2901942917.
py:1: FutureWarning: 'BM' is deprecated and will be removed in a future vers
ion, please use 'BME' instead.
monthly = df.asfreq('BM')
```

In [127... `monthly.head()`

Out[127...

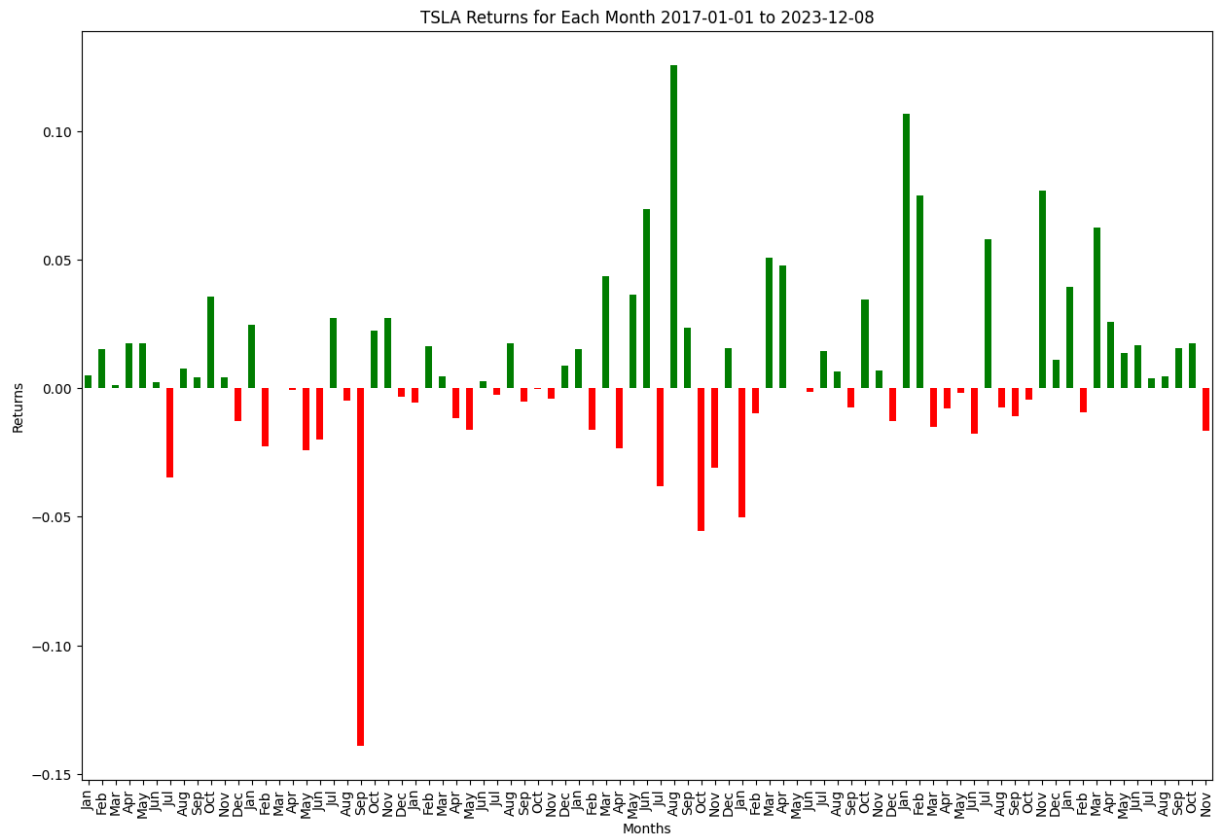
	Price	Date	Close	High	Low	Open	Volume	Returns	M
0		2017-01-31	16.795334	17.059334	16.513332	16.615999	61741500.0	0.005187	
1		2017-02-28	16.666000	16.733334	16.260000	16.279333	91171500.0	0.015270	
2		2017-03-31	18.553333	18.645332	18.421333	18.582001	49419000.0	0.001367	
3		2017-04-28	20.938000	20.986668	20.533333	20.655333	67582500.0	0.017626	
4		2017-05-31	22.733999	22.859333	22.344000	22.512667	149451000.0	0.017636	

In [128...

```
monthly['ReturnsPositive'] = 0 < monthly['Returns']
monthly['Date'] = pd.to_datetime(monthly['Date'])
monthly['Date'] = monthly['Date'].apply(mdates.date2num)
```

In [129...

```
colors = monthly>ReturnsPositive'.map({True: 'g', False: 'r'})
monthly['Returns'].plot(kind='bar', color = colors, figsize=(15,10))
plt.xlabel("Months")
plt.ylabel("Returns")
plt.title(stock + " Returns for Each Month " + str(start) + ' to ' + str(end))
plt.xticks(monthly.index, monthly['Month_Name'])
plt.show()
```



In []:

Yearly returns

```
In [130... yearly = df.asfreq('BY')
yearly['Returns'] = df['Close'].pct_change().dropna()
yearly = yearly.reset_index()
yearly['Years'] = yearly['Date'].dt.year
print(yearly)
```

	Price	Date	Close	High	Low	Open	Volume
0		2017-12-29	20.756666	21.094000	20.666668	21.078667	56658000
1		2018-12-31	22.186666	22.614000	21.684000	22.519333	94534500
2		2019-12-31	27.888666	28.086000	26.805332	27.000000	154285500
3		2020-12-31	235.223328	239.573334	230.373337	233.330002	148949700
4		2021-12-31	352.260010	360.666656	351.529999	357.813324	40733700
5		2022-12-30	123.180000	124.480003	119.750000	119.949997	157777300

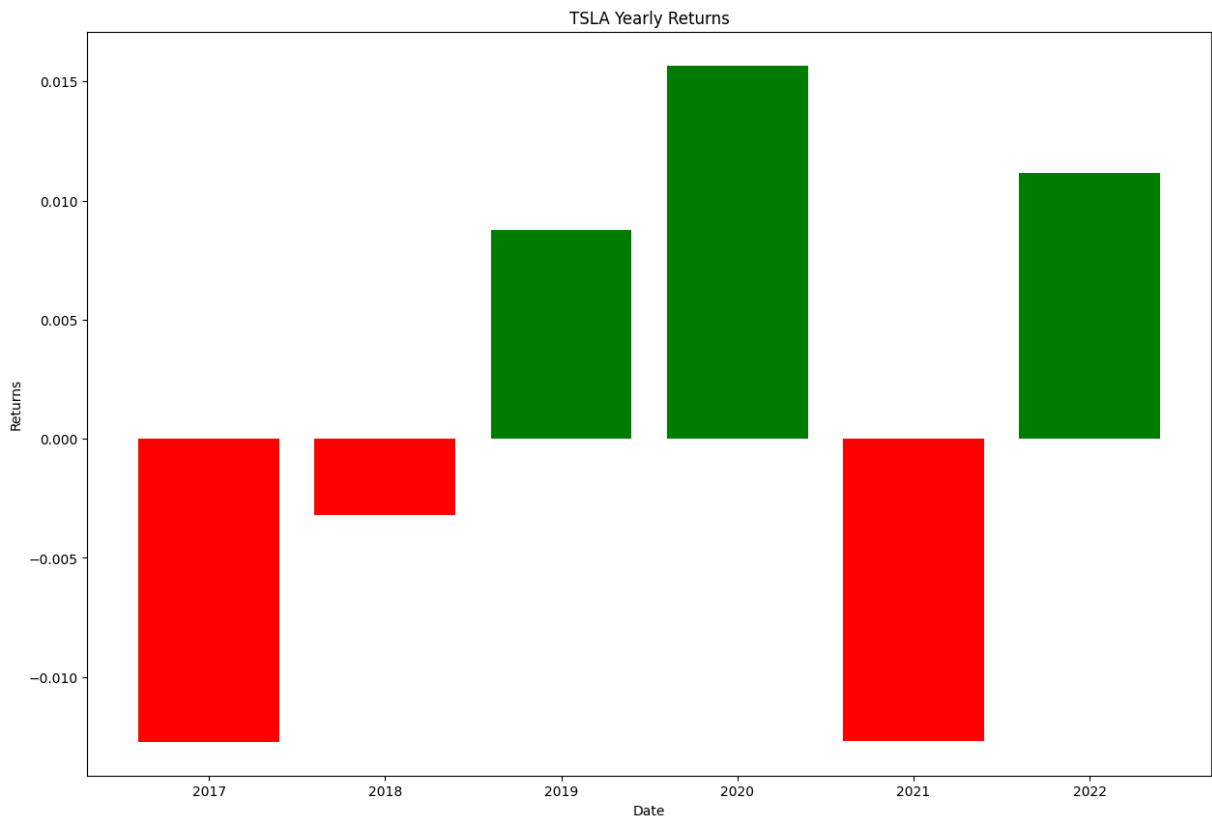
	Price	Returns	Years
0		-0.012716	2017
1		-0.003205	2018
2		0.008753	2019
3		0.015674	2020
4		-0.012669	2021
5		0.011164	2022

```
/var/folders/g7/8fm0cc6d7p13pn53xrhqdw5c0000gn/T/ipykernel_33896/3878586787.py:1: FutureWarning: 'BY' is deprecated and will be removed in a future version, please use 'BYE' instead.  
yearly = df.asfreq('BY')
```

```
In [131... yearly['ReturnsPositive'] = 0 < yearly['Returns']  
yearly['Date'] = pd.to_datetime(yearly['Date'])  
yearly['Date'] = yearly['Date'].apply(mdates.date2num)
```

```
In [132... colors = yearly>ReturnsPositive'.map({True: 'g', False: 'r'})  
plt.gcf()  
plt.gcf()  
plt.figure(figsize=(15,10))  
plt.bar(yearly['Years'], yearly['Returns'], color=colors, align='center')  
plt.title(stock + ' Yearly Returns')  
plt.xlabel('Date')  
plt.ylabel('Returns')  
plt.show()
```

<Figure size 640x480 with 0 Axes>



Yearly Returns Avg

```
In [133... df['Returns'] = df['Close'].pct_change().dropna()  
yearly_returns_avg = df['Returns'].groupby([df.index.year]).mean()  
print(yearly_returns_avg)
```

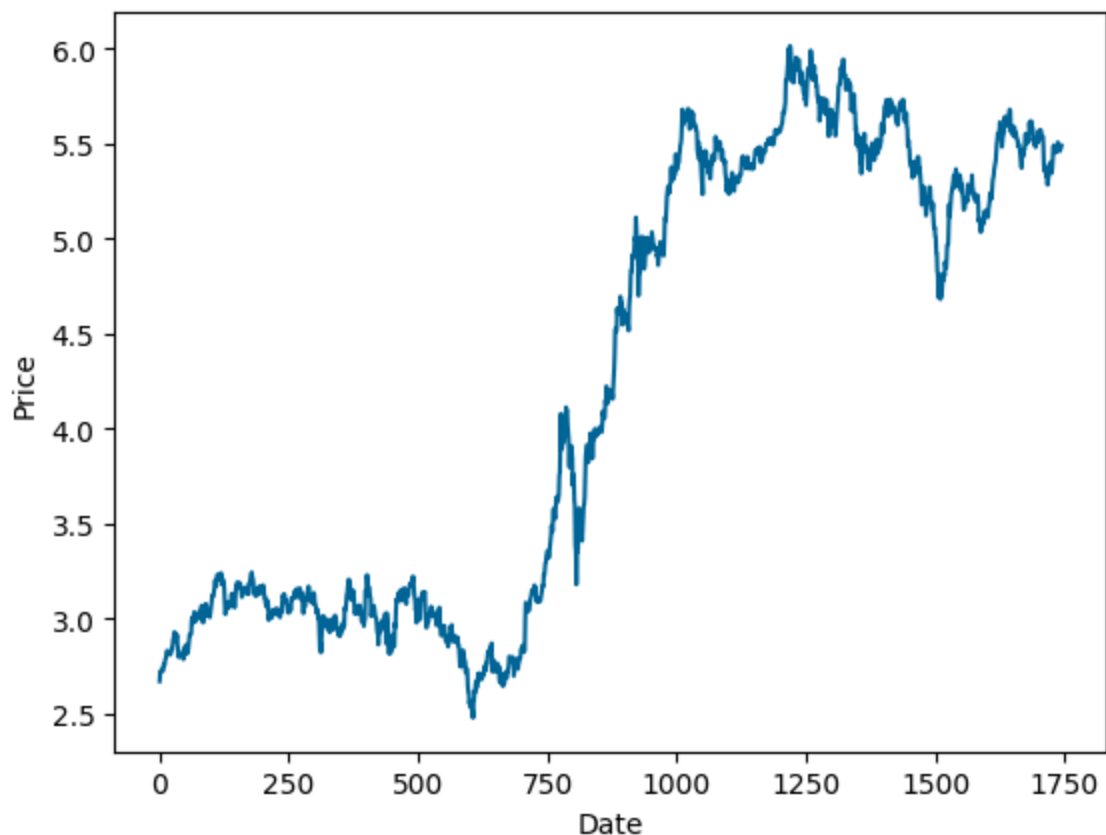
```
Date
2017    0.001693
2018    0.000932
2019    0.001391
2020    0.010047
2021    0.002188
2022   -0.003293
2023    0.003486
Name: Returns, dtype: float64
```

```
In [ ]:
```

```
In [134... df = df.reset_index().rename(columns={'Date':'ds', 'Adj Close':'y'})
#df['ds'] = pd.to_datetime(df.index)
#df['y'] = pd.DataFrame(df['Adj Close'])
```

```
In [135... # Log Transform Data
#df['y'] = pd.DataFrame(np.log(df['y']))
df['y'] = np.log(df['Close'])

# plot data
ax = df['y'].plot(color='#006699');
ax.set_ylabel('Price');
ax.set_xlabel('Date');
plt.show()
```



Data processings : creating training and testing data

```
In [136... # split data into train and test
df_train = df[:740]
df_test = df[740:]
```

```
In [137... df_train
```

```
Out[137... Price      ds      Close      High      Low      Open      Volume      Returns
2017-
0  01-03  14.466000  14.688667  14.064000  14.324000  88849500      NaN  2.
2017-
1  01-04  15.132667  15.200000  14.287333  14.316667  168202500  0.046085  2.
2017-
2  01-05  15.116667  15.165333  14.796667  15.094667  88675500  -0.001057  2.
2017-
3  01-06  15.267333  15.354000  15.030000  15.128667  82918500  0.009967  2.
2017-
4  01-09  15.418667  15.461333  15.200000  15.264667  59692500  0.009912  2.
...
2019-
735 12-04  22.202000  22.524000  22.190001  22.516666  82995000  -0.009429  3.
2019-
736 12-05  22.024668  22.294666  21.816668  22.188667  55869000  -0.007987  3.
2019-
737 12-06  22.392668  22.590668  22.318001  22.333332  114186000  0.016709  3.
2019-
738 12-09  22.635332  22.963333  22.338667  22.439333  135346500  0.010837  3.
2019-
739 12-10  23.256001  23.382000  22.620667  22.664000  132424500  0.027420  3.
```

740 rows × 8 columns

```
In [138... df_test
```

Out [138...

Price	ds	Close	High	Low	Open	Volume	Returr
740	2019-12-11	23.513332	23.812668	23.406000	23.458668	103467000	0.01106
741	2019-12-12	23.978666	24.182667	23.548668	23.661333	116458500	0.01979
742	2019-12-13	23.892668	24.347334	23.642668	24.070000	98563500	-0.00358
743	2019-12-16	25.433332	25.573999	24.166668	24.170000	272613000	0.06448
744	2019-12-17	25.266001	25.700001	25.059999	25.266001	127452000	-0.00657
...	...	...	...	...	...	...	...
1740	2023-12-01	238.830002	240.190002	231.899994	233.139999	121331700	-0.00520
1741	2023-12-04	235.580002	239.369995	233.289993	235.750000	104099800	-0.01360
1742	2023-12-05	238.720001	246.660004	233.699997	233.869995	137971100	0.01332
1743	2023-12-06	239.369995	246.570007	239.169998	242.919998	126436200	0.00272
1744	2023-12-07	242.639999	244.080002	236.979996	241.550003	107142300	0.01366

1005 rows x 8 columns

Model Fitting with train data

Ref:

https://facebook.github.io/prophet/docs/uncertainty_intervals.html

Description

Make dataframe with future dates for forecasting.

Usage

- `make_future_dataframe(m, periods, freq = "day", include_history = TRUE)`

Arguments

`m` : Prophet model object. `periods` : Int number of periods to forecast forward.

freq : 'day', 'week', 'month', 'quarter', 'year', 1(1 sec), 60(1 minute) or 3600(1 hour).

include_history : Boolean to include the historical dates in the data frame for predictions.

```
In [139... # Model Fitting
# instantiate the Prophet class
mdl = Prophet(interval_width=0.95, daily_seasonality=True)

# fit the model on the training data
mdl.fit(df_train)

# define future time frame
future = mdl.make_future_dataframe(periods=24, freq='MS') # periods 24 months
```

```
16:21:02 - cmdstanpy - INFO - Chain [1] start processing
16:21:02 - cmdstanpy - INFO - Chain [1] done processing
```

```
In [140... # generate the forecast
forecast = mdl.predict(future)
forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail()
```

```
Out[140...      ds      yhat  yhat_lower  yhat_upper
759 2021-08-01  3.662812    0.247383    6.907423
760 2021-09-01  3.622936    0.041016    7.084059
761 2021-10-01  3.692690   -0.189327    7.474751
762 2021-11-01  3.902022   -0.243755    7.927403
763 2021-12-01  4.013681   -0.363028    8.246071
```

```
In [141... forecast['yhat_lower'].head()
```

```
Out[141... 0    2.551589
1    2.563238
2    2.577682
3    2.572278
4    2.606983
Name: yhat_lower, dtype: float64
```

```
In [142... forecast
```

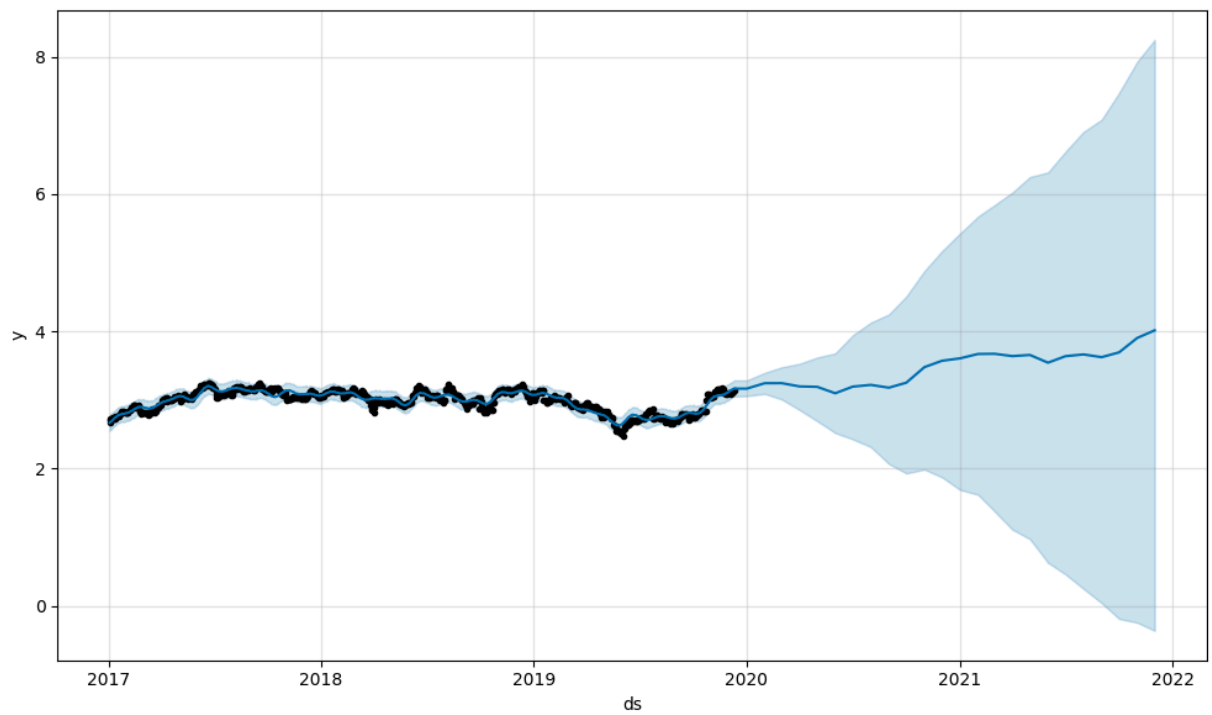
Out [142...

	ds	trend	yhat_lower	yhat_upper	trend_lower	trend_upper	additive_te
0	2017-01-03	2.637354	2.551589	2.779121	2.637354	2.637354	0.029
1	2017-01-04	2.641595	2.563238	2.793217	2.641595	2.641595	0.033
2	2017-01-05	2.645837	2.577682	2.805312	2.645837	2.645837	0.035
3	2017-01-06	2.650078	2.572278	2.797580	2.650078	2.650078	0.036
4	2017-01-09	2.662802	2.606983	2.824679	2.662802	2.662802	0.050
...	...	...	...	...	...	...	...
759	2021-08-01	3.843421	0.247383	6.907423	0.474626	7.073890	-0.180
760	2021-09-01	3.881140	0.041016	7.084059	0.243479	7.367746	-0.258
761	2021-10-01	3.917641	-0.189327	7.474751	0.026304	7.671429	-0.224
762	2021-11-01	3.955359	-0.243755	7.927403	-0.191521	7.959510	-0.053
763	2021-12-01	3.991861	-0.363028	8.246071	-0.378322	8.229343	0.021

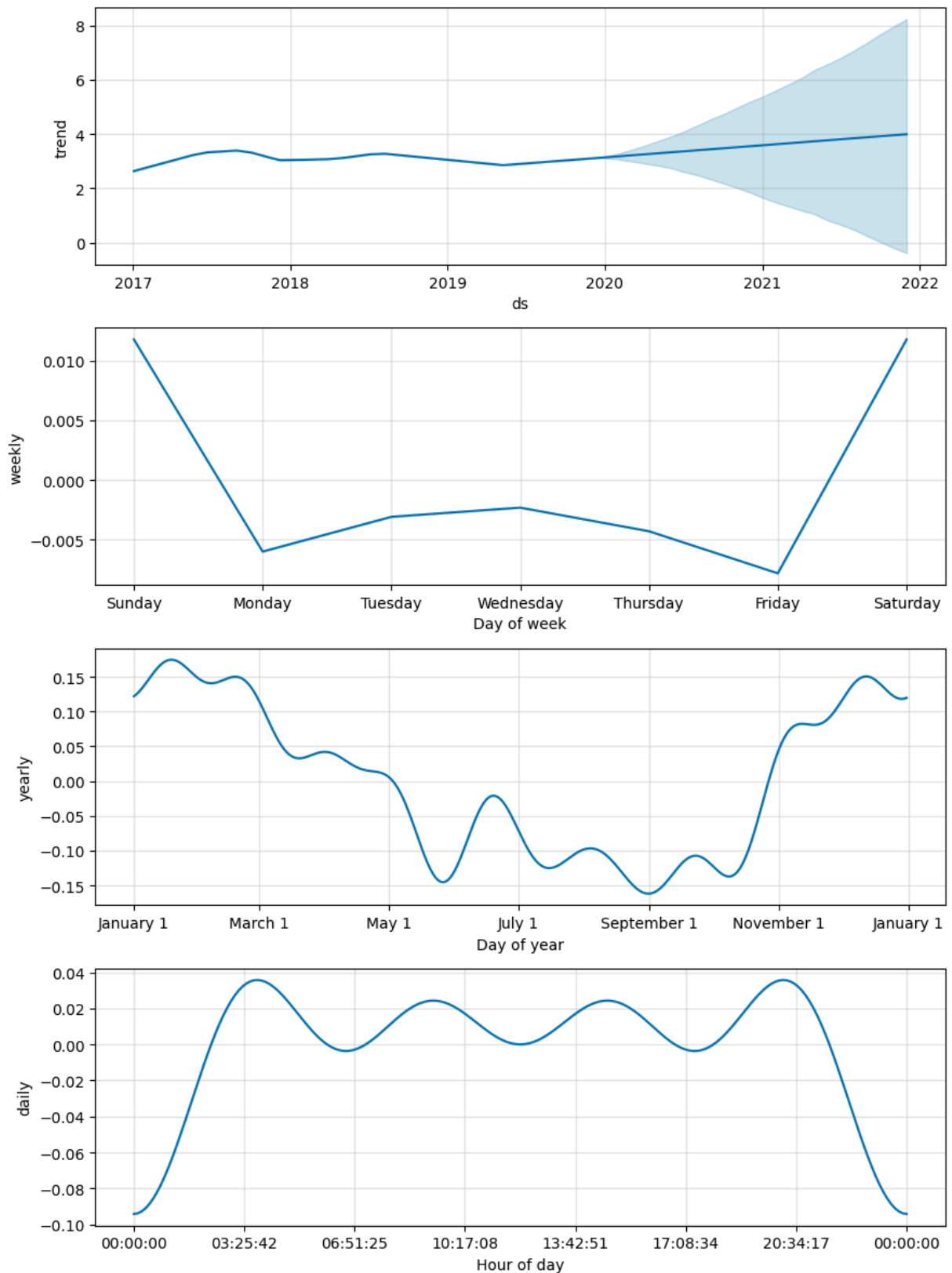
764 rows × 22 columns

In [143...

```
mdl.plot(forecast);  
plt.show()
```



```
In [144... # plot time series components  
mdl.plot_components(forecast)  
plt.show()
```



```
In [145... # retransform using e
y_hat = np.exp(forecast['yhat'][:740])
y_true = np.exp(df['y'])

# compute the mean square error
mse = ((y_hat - y_true) ** 2).mean()
# mse = ((forecast['yhat'][:740] - df['y']) ** 2).mean()
```

```
print('Prediction quality: {:.2f} MSE ({:.2f} RMSE)'.format(mse, math.sqrt(mse)))
```

Prediction quality: 1.31 MSE (1.14 RMSE)

```
In [146... se = np.square(forecast.loc[:739, 'yhat'] - df['y'])
mse = np.mean(se)
rmse = np.sqrt(mse)
rmse
```

Out[146... np.float64(0.05779008483138419)

```
In [147... #mae = mean_absolute_error(forecast.loc[:739, 'yhat'], df['y'])
#print('MAE: %.3f' % mae)

mae = mean_absolute_error(df.loc[:739, 'y'], forecast.loc[:739, 'yhat'])
print('MAE: %.3f' % mae)
```

MAE: 0.046

```
In [148... df['y']
```

```
Out[148... 0      2.671801
1      2.716856
2      2.715798
3      2.725715
4      2.735579
...
1740    5.475752
1741    5.462051
1742    5.475291
1743    5.478010
1744    5.491579
Name: y, Length: 1745, dtype: float64
```

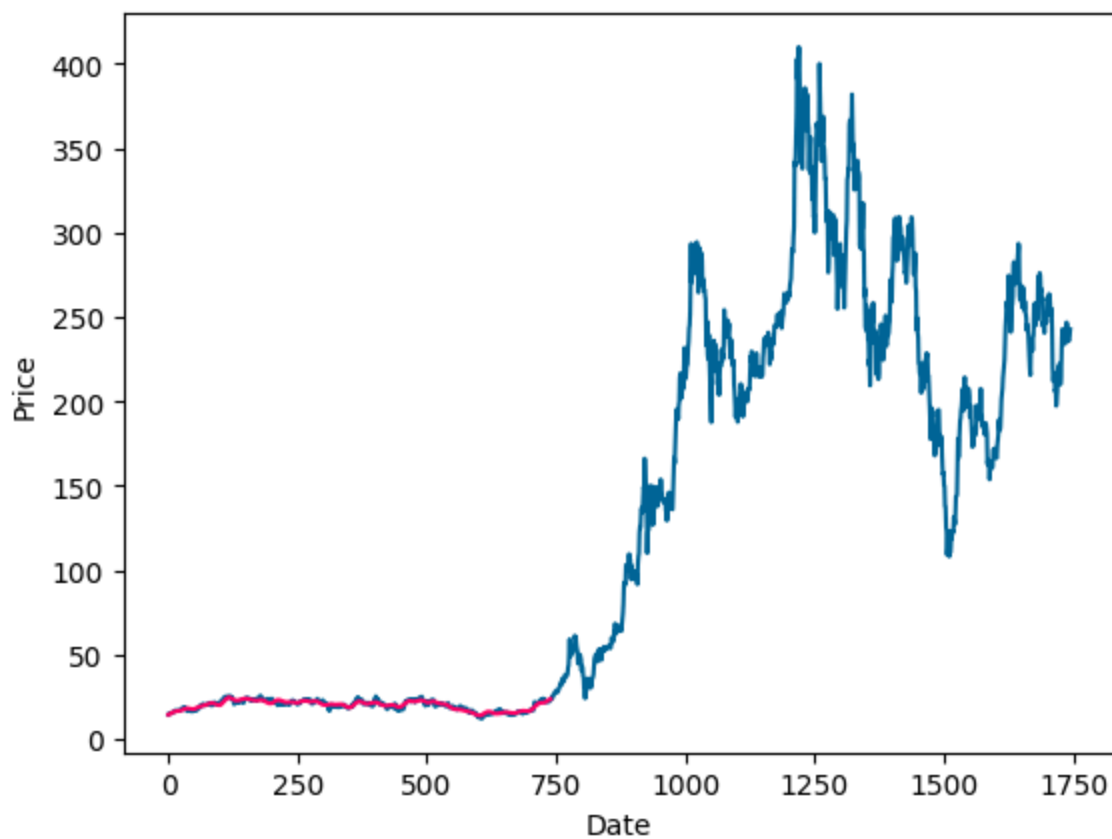
```
In [149... forecast['yhat'][:740]
```

```
Out[149... 0      2.667264
1      2.675574
2      2.681495
3      2.686148
4      2.713238
...
735    3.139419
736    3.142892
737    3.144477
738    3.158823
739    3.164794
Name: yhat, Length: 740, dtype: float64
```

```
In [150... forecast.loc[:739, 'yhat']
```

```
Out[150...] 0      2.667264
            1      2.675574
            2      2.681495
            3      2.686148
            4      2.713238
            ...
            735     3.139419
            736     3.142892
            737     3.144477
            738     3.158823
            739     3.164794
Name: yhat, Length: 740, dtype: float64
```

```
In [151...] plt.plot(y_true, label='Original', color='#006699');
plt.plot(y_hat, color='#ff0066', label='Forecast');
plt.ylabel('Price');
plt.xlabel('Date');
plt.show()
```



```
In [152...] import plotly.io as pio
pio.renderers.default = 'iframe'
```

```
In [153...] plot_plotly mdl, forecast)
```



```
In [70]: !jupyter nbconvert --to webpdf "Forecasting using Time Series Analysis.ipynb"
```

```
[NbConvertApp] Converting notebook Forecasting using Time Series Analysis.ipynb to webpdf
```

```
[NbConvertApp] WARNING | Alternative text is missing on 11 image(s).
```

```
[NbConvertApp] Building PDF
```

```
[NbConvertApp] PDF successfully created
```

```
[NbConvertApp] Writing 715722 bytes to Forecasting using Time Series Analysis.pdf
```

```
In [ ]:
```